

Satellite and Drone-Based KPI Framework for Tea Plantation Intelligence

Crop Health & Disease Intelligence — Estate Owner Briefing

Section A of the Satellite & Drone KPI Framework

KPI	Question It Answers for You	What It Means	Why It Matters / Alert Levels
TOP-LINE INDICATOR			
Block Health Score	Which blocks need my attention this week?	A single 0–100 score per block combining canopy strength, canopy structure, and disease signals into one number — the top-line indicator to check first.	Good >80 • Watch 60–80 • Act <60
GROWTH & TIMING			
Crop Vigor Score	Is yield at risk before it shows up on the weighing scale?	Tracks canopy greenness and strength from satellite imagery, giving an early warning of decline before it's visible on the ground.	Good: strong greenness • Watch: below normal • Act: well below normal
Flush Readiness Score	Am I plucking at the right time?	Signals when new growth is sufficient for plucking. Picking too early or late costs both yield and leaf quality.	Ready • Approaching • Weak flush / delayed
Canopy Density Score	Which blocks need pruning or replanting?	Measures canopy thickness and structure — a long-term driver of yield potential and the underlying value of the estate's tea bushes.	Stable/growing • Declining • >15% drop vs. baseline
ENVIRONMENTAL STRESS DIAGNOSTICS			
Moisture Stress Score	Is my crop water-stressed even though it still looks green?	Measures canopy water content directly, catching drought or irrigation gaps before leaves visibly wilt. Usually the earliest of the four stress signals to move.	Good: adequate moisture • Watch: declining • Act: significant deficit
Nitrogen Stress Score	Are my bushes short on nitrogens, and will the next flush be lighter than expected?	Reads chlorophyll/nitrogen status in the canopy, flagging fertilizer gaps before leaf yellowing is visible to a field walk.	Good: sufficient • Watch: declining • Act: deficient
Heat Stress Score	Is a block running hotter than it should during dry, still weather?	Compares canopy surface temperature to expected levels. Sustained heat stress slows flush and makes plants more vulnerable to disease and pests.	Good: normal range • Watch: elevated • Act: sustained high heat
DISEASE RISK & SPREAD			

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Disease Risk Score	Where should I spray preventively, before an outbreak starts?	A forward-looking probability of disease developing, combining plant condition with weather patterns — a forecast, not a current count.	Low <20% • Moderate 20–50% • High >50%
Disease Hotspot Coverage	How large is a current outbreak, and what will treatment cost?	The percentage of a block currently showing disease symptoms — the actual scale of a problem happening now, distinct from the forecast above.	Low <5% • Watch 5–15% • Act >15%
Disease Spread rate	How quickly is disease spreading across the estate?	Measures the rate at which disease-affected area is increasing over time, showing whether an outbreak is contained, stable, or expanding.	Target: Stable or declining spread • Watch: Increasing affected area • Act: Rapid week-to-week expansion. Rapid spread indicates current control measures are insufficient and immediate intervention is required.

Input & Resource Utilization — Estate Owner Briefing

Section B of the Satellite & Drone KPI Framework — 4 KPIs organized into 3 groups

KPI	Question It Answers for You	What It Means	Why It Matters / Alert Levels
NUTRIENT MANAGEMENT			
Nitrogen Use Reduction	Are we spending less on Nitrogen with early detection?	Measures the cut in spray volume achieved by targeting only nitrogen deficiency zones instead of blanket application across the estate — a direct, trackable cost-saving number for ownership.	Good <5% of block • Watch: rising • Act: significant area affected
CHEMICAL INPUT MANAGEMENT			
Agrochemical Use Reduction	Are we spending less on pesticide and fungicide without more disease?	Measures the cut in spray volume achieved by targeting only disease-mapped zones instead of blanket spraying across the estate — a direct, trackable cost-saving number for ownership.	Target: 10–20% reduction vs. baseline, without rising disease incidence

Yield & Financial Metrics — Estate Owner Briefing

KPI	Question It Answers for You	What It Means	Why It Matters / Alert Levels
YIELD OUTLOOK			
Yield Anomaly Index	Are we ahead of or behind a normal season right now?	The percentage gap between current and expected yield — the 'how worried should I be' number, distinct from the forecast above because it's a relative comparison rather than an absolute volume.	Good ≥0% vs. baseline • Watch -5% to -10% • Act < -10%
FINANCIAL IMPACT DRIVERS			
Revenue Protected	How much money did early disease detection actually save me?	Estimated value of crop saved by catching disease early via Section A's disease KPIs — turns an agronomy alert into a rupee figure.	Target: positive and increasing over time
Revenue at Risk	How much revenue is exposed right now because of drought or crop stress?	Translates moisture/stress readings into a financial exposure number. Tea drought research shows 7.72–12.52% yield loss during drought periods, so blocks flagged critical on moisture should be treated as financial-risk zones.	Target: 0 or decreasing
Input Cost Savings	Is targeted spraying and fertilizing actually saving money over blanket application?	Directly measures cost reduction from precision application versus treating the whole estate uniformly.	Target: positive savings each quarter; report in LKR/ha and % vs. baseline

Soil Health Parameters — Estate Owner Briefing

KPI	Question It Answers for You	What It Means	Why It Matters / Alert Levels
LAND DEGRADATION RISK			
Erosion Risk Area	Which parts of my estate are losing topsoil?	Maps exposed bare-soil areas vulnerable to erosion. Protecting soil now avoids expensive rehabilitation and yield loss later.	Good <5% of block • Watch 5–15% • Act >15%
Soil Vulnerability Classification	Which areas need soil rehabilitation investment, and where should I prioritize it?	Classifies degraded or exposed zones by mineral composition, flagging where the soil itself — not just the crop — needs attention. A capital-planning input rather than a week-to-week score.	Target: no expansion of high-vulnerability zones (read as a map; thresholds vary by soil type)
DRAINAGE RISK			
Waterlogging Risk	Where is poor drainage threatening root health?	Identifies standing water and poor drainage — a distinct risk from drought, protecting against root disease and yield loss caused by too much water rather than too little.	Good: no standing-water signal • Watch: mild signal • Act: strong water presence

Predictive & AI Trend Indicators — Estate Owner Briefing

KPI	Question It Answers for You	What It Means	Why It Matters / Alert Levels
30 day Yield Forecast	<i>How much green leaf should I expect to harvest, and how does that compare to a normal block?</i>	Predicted green-leaf yield for the coming month, calibrated against each block's own historical baseline. This folds in the original framework's per-hectare 'Green Leaf Yield Estimate' — same underlying number, so it isn't shown to the owner twice.	Target: forecast error $\leq 10\text{--}15\%$ after calibration; benchmark block-by-block, not one universal figure
14-Day Disease Risk Probability	<i>Which blocks are most likely to experience a disease outbreak in the next two weeks?</i>	<i>Predicts the probability of disease developing in each block by combining vegetation condition, moisture status, weather patterns, and historical disease trends.</i>	Target: $<20\%$ • Watch: $20\text{--}50\%$ • Act: $>50\%$. Enables preventive spraying before disease becomes visible, reducing crop losses and treatment costs.
Block Priority Score	<i>Which blocks should I inspect and manage first?</i>	<i>An AI-generated ranking of plantation blocks based on overall crop condition, disease risk, moisture stress, nutrient status, and expected yield impact.</i>	Target: No block should remain in High Priority for more than two monitoring cycles without intervention. Helps managers allocate labour and resources where they will have the greatest impact.

Asset Monitoring Modules using AI

The land and asset requirements are delivered as maps, inventories and surveys — refreshed annually or on demand rather than watched every month.

KPI	Question It Answers for You	What It Means	Why It Matters / Alert Levels
Estate Infrastructure Survey & Map	<i>What do my roads, buildings, and boundaries look like today?</i>	<i>Shows an up-to-date map of roads, buildings, drainage, and estate boundaries.</i>	Target: Complete survey at least once a year. Missing or damaged infrastructure should be inspected and repaired.
Estate Tree Inventory – Shade & Fuel Wood	<i>How many shade and fuel-wood trees do I have, and where are they?</i>	<i>Counts and maps shade trees and fuel-wood trees across the estate.</i>	Target: Keep tree numbers stable or increasing each year. Falling tree numbers mean more planting may be needed.
Natural Habitat Mapping	<i>Where are the natural areas that should be protected?</i>	<i>Maps forests, streams, and other natural areas inside or around the estate.</i>	Target: No loss of natural habitat each year. Loss of habitat should lead to protection or restoration work.